



João Luís Pereira da Silva Marques

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Gender: Male

ABOUT ME

<https://www.linkedin.com/in/joaoluis-silvamarques/>
<https://github.com/jm651120>

Recently graduated Master in Data Science and Advanced Analytics with a solid background in Computer Engineering.
Currently working as a Data & Analytics Associate, developing and implementing complex metrics for financial transaction analysis. Passionate about utilizing advanced machine learning techniques, big data, and automated workflows to extract actionable business insights and drive data-informed decision-making.

WORK EXPERIENCE

Avanade

City: Lisbon | **Country:** Portugal

[01/06/2026 – Current] **Data Engineering Associate**

- Developed and maintained Python and SQL-based ETL pipelines within an Azure Databricks Data Lake environment, managing complex data flows across ODS and SAD tables for the banking sector.
- Engineered customized data metrics to analyze financial transaction patterns, creating dynamic aggregations for KPIs such as 30-day cash withdrawal summaries and volume thresholds for bank transfers.
- Developed and modified ETL notebooks dedicated to Row-Level Security (RLS) management, ensuring strict user and department access control.
- Supported business requirements analysis by drafting technical solution proposals (DPD) for Risk model access requests.
- Applied development best practices, technical documentation, and version control using Git/GitHub.

NOVA Information Management School

City: Lisbon | **Country:** Portugal

[09/09/2025 – 26/06/2026] **Machine Learning Researcher (MSc Thesis)**

- Evaluated the financial impact of Energy Performance Certificates (EPC) on UK residential properties using a multi-source dataset of 514,345 transactions (2021-2025).
- Developed an automated valuation model using advanced ensemble learning algorithms (XGBoost, LightGBM, Random Forest) to capture non-linear spatial and structural dynamics.
- Achieved an R^2 of 0.8249 and a Mean Absolute Error of £60,868 with the champion XGBoost model.
- Implemented the SHAP framework to explicitly quantify the market's environmental margin, identifying a "Green Premium" of +£13,506 for Band A properties and a "Brown Discount" of -£12,017 for Band G properties.

EDUCATION AND TRAINING

[09/09/2024 – 26/06/2026] **Master's in Data Science and Advanced Analytics**

Nova Information Management School <https://www.novaims.unl.pt/>

City: Lisboa | **Country:** Portugal | **Field(s) of study:** informat

[27/09/2021 – 19/07/2024] **Bachelor's in Computer Engineering**

NOVA School of Science and Technology <https://www.fct.unl.pt/>

City: Almada | **Country:** Portugal

LANGUAGE SKILLS

Mother tongue(s): Português

Other language(s):

Inglês

LISTENING C1 READING C1 WRITING C1

SPOKEN PRODUCTION B2 SPOKEN INTERACTION C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Python (Pandas, NumPy, Scikit-Learn, Matplotlib) | Advanced Predictive Modeling & Ensemble Learning | Explainable AI (SHAP) & CRISP-ML Pipeline | SQL (MySQL, SQL Server) & Git/GitHub | NLP & Text Analytics | LLM & RAG | Java, C/C++ | HTML/CSS/JS | Azure Databricks (PySpark, Notebooks, ETL, RLS) | Web Scraping (Firecrawl, BeautifulSoup)

PROJECTS

[12/06/2026]

FitBot - AI Assistant for FIT IT Gym

- Developed a production-grade Retrieval-Augmented Generation (RAG) chatbot using Python, LangChain, and Streamlit to provide 24/7 grounded answers for a commercial gym.
- Engineered an automated data pipeline with Firecrawl and APScheduler for weekly schedule updates and monthly full-site re-crawls, utilizing MD5 content hashing for change detection.
- Implemented intent-routed semantic search using ChromaDB and 100% local Sentence-Transformers embeddings.
- Designed a self-healing LLM architecture with an automatic model fallback chain (Llama 3.3 70B to 8B via Groq) and conversational memory rewriting for seamless multi-turn interactions.
- Created custom parsing logic to extract structured schedule data directly from embedded JSON booking widgets, strictly enforcing a zero-hallucination policy.

Link: <https://fitit-chatbot.streamlit.app/>

CERTIFICATIONS

[McKinsey & Company, 23/06/2026]

McKinsey Forward

Mode of learning: Online

[Google, 15/02/2024]

Foundations: Data, Data, Everywhere

Mode of learning: Online

[Cambridge University, 18/07/2019]

Cambridge Assessment

Mode of learning: Presential

VOLUNTEERING

[01/02/2026 – 09/02/2026]

Missão País Meia Via

A Catholic university volunteer movement in Portugal where students participate in week-long missions in rural communities, engaging in social support activities, community outreach, and faith-based service with local institutions such as schools, nursing homes, and charities.

2026: Conducted door-to-door outreach to assist local residents and organized recreational activities at a daycare, a rehabilitation center for people with disabilities, and a nursing home.

2025: Co-organized a community theater play that served as a food drive for the Cruz Vermelha of Santarém.

[01/07/2018 – 01/08/2018]

MOVE UP Project Leiria - Portugal

Participated in the Move Up project, an initiative aimed at developing participants' social and communication skills. Together with three colleagues, I was assigned a group of children and organized weekly recreational activities to promote interaction, teamwork, and personal development.